

Rice University  
Department of Civil and Environmental Engineering  
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James Doss-Gollin

## James Doss-Gollin

### RESEARCH INTERESTS

My research builds probabilistic hazard models that enable climate risk analysis and bottom-up decision support for complex infrastructure systems, with a focus on urban flooding. I integrate Earth science, artificial intelligence, and decision analysis to examine non-stationary hazards, assess their effects on built environments, and develop adaptive risk-management approaches.

### APPOINTMENTS

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| <b>Rice University</b>                                                                                |       |
| Assistant Professor, Department of Civil & Environmental Engineering                                  | 2021– |
| Affiliated Faculty, Severe Storm Prediction, Education, and Evacuation from Disasters (SSPEED) Center | 2021– |
| Affiliated Faculty, Ken Kennedy Institute                                                             | 2023– |
| <b>The Pennsylvania State University</b>                                                              |       |
| Postdoctoral Scholar, Earth & Environmental Systems Institute                                         | 2020  |

### EDUCATION

|                                            |      |
|--------------------------------------------|------|
| <b>Columbia University</b>                 |      |
| Ph.D. in Earth & Environmental Engineering | 2020 |
| M.S. in Earth & Environmental Engineering  | 2016 |
| <b>Yale University</b>                     |      |
| B.S. in Mechanical Engineering             | 2015 |

### AWARDS AND HONORS

|                                                                                                                                |      |
|--------------------------------------------------------------------------------------------------------------------------------|------|
| <b>NSF CAREER Award</b> , National Science Foundation                                                                          | 2026 |
| <b>Outstanding Reviewer Award</b> , Earth's Future                                                                             | 2023 |
| <b>Nickolas and Liliana Themelis Fellowship</b> , Fu Foundation School of Engineering and Applied Science, Columbia University | 2018 |
| <b>Graduate Research Fellowship, Climate and Large-Scale Atmospheric Dynamics</b> , National Science Foundation                | 2017 |
| <b>Presidential Distinguished Fellowship</b> , Fu Foundation School of Engineering and Applied Science, Columbia University    | 2015 |
| <b>Distinction in Major</b> , Department of Mechanical Engineering and Materials Science, Yale University                      | 2015 |
| <b>Legacy Award</b> , New Haven Promise                                                                                        | 2015 |
| <b>Larry Coben '79 Fellowship</b> , Yale University                                                                            | 2014 |
| <b>Vance-Carter Travel Award</b> , Yale University                                                                             | 2013 |
| <b>Thomas C. Barry Travel Award</b> , Yale University                                                                          | 2012 |

### ADVISEE AWARDS

|                                                                                                       |      |
|-------------------------------------------------------------------------------------------------------|------|
| <b>Dongwook Kim</b> : Karen and John Huff Graduate Fellowship in Civil and Environmental Engineering. | 2025 |
| <b>Yuchen Lu</b> : H.W. Reeves Endowed Scholarship.                                                   | 2022 |

## PUBLICATIONS

† denotes Rice advisee publication. Google Scholar citations: 759, h-index: 13, i10-index: 14.

### IN PREP. / UNDER REVIEW

- [3] Baer, J. A., Sebastian, A., Grimley, L. E., **Doss-Gollin, J.**, Wright, D. B., Hussain, M. A., and Webber, M. : *Neglecting Spatiotemporal Rainfall Variability Underestimates Flood Hazard and Risk.* . DOI: [10.21203/rs.3.rs-8593870/v1](https://doi.org/10.21203/rs.3.rs-8593870/v1) —
- [2] Hancock, C. L., Dee, S. G., Haider, M. R., **Doss-Gollin, J.**, Lehner, F., Murphy, K., and Munoz, S. E. : *Robust 21st Century Hydrological Trends in the Mississippi River Basin from CMIP6: West-Gets-Drier, East-Gets-Wetter.* —
- [1] Pollack, A., Benedict, J., Deb, M., **Doss-Gollin, J.**, Judi, D., Lehman, W., Lutz, N., Reesman, C., Sarazen, E., Son, Y., Srikrishnan, V., Sun, N., and Keller, K. : *Unrefined National Building Inventories Can Mislead Risk Assessments and Decisions.* . DOI: [10.2139/ssrn.5575271](https://doi.org/10.2139/ssrn.5575271) —

### JOURNAL ARTICLES

- [26] Murphy, K., Dee, S., Hancock, C., Pitchon, E., **Doss-Gollin, J.**, Wallace, E., and Muñoz, S. : “The Changing Relationship between the Bermuda High, Great Plains Low-Level Jet, and Mississippi River Basin Hydroclimate from the Last Millennium to 2100”. *Journal of Geophysical Research: Atmospheres*. DOI: [10.1029/2025JD044340](https://doi.org/10.1029/2025JD044340) 2026
- [25] O'Donnell, M., Dee, S., **Doss-Gollin, J.**, and Muñoz, S. E. : “Hydrologic Whiplash in the Mississippi River Basin: Mechanisms and Projections”. *Geophysical Research Letters*. DOI: [10.1029/2026GL122807](https://doi.org/10.1029/2026GL122807) 2026
- [24] Pollack, A. B., Auermuller, L., Burleyson, C. D., Campbell, J., Condon, M., Cooper, C., Coronese, M., Dangendorf, S., **Doss-Gollin, J.**, Hegde, P., Helgeson, C., Kopp, R. E., Kwakkel, J., Lesk, C., Mankin, J., Nicholas, R. E., Rice, J., Roth, S., Srikrishnan, V., Scheeler, M., Tuana, N., Vernon, C., Zhao, M., and Keller, K. : “Unlocking the Benefits of Transparent and Reusable Science for Climate Risk Management”. *Proceedings of the National Academy of Sciences*. DOI: [10.1073/pnas.2422157123](https://doi.org/10.1073/pnas.2422157123) 2026
- [23] Haider, M. R., Dee, S., **Doss-Gollin, J.**, Dunne, K., and Muñoz, S. : “Impact of 21st Century Climate Change on Mississippi River Basin Discharge in CESM2 Large Ensemble Projections”. *Global and Planetary Change*. DOI: [10.1016/j.gloplacha.2025.104742](https://doi.org/10.1016/j.gloplacha.2025.104742) 2025
- [22] Liu, C., Kowal, D. R., **Doss-Gollin, J.**, and Vannucci, M. : “Bayesian Functional Graphical Models with Change-Point Detection”. *Computational Statistics and Data Analysis*. DOI: [10.1016/j.csda.2024.108122](https://doi.org/10.1016/j.csda.2024.108122) 2025
- [21] Liu, Y., **Doss-Gollin, J.**, Dai, Q., Veeraraghavan, A., and Balakrishnan, G. : “Downscaling Extreme Precipitation with Wasserstein Regularized Diffusion”. *IEEE Transactions on Geoscience and Remote Sensing*. DOI: [10.1109/TGRS.2025.3611872](https://doi.org/10.1109/TGRS.2025.3611872) 2025
- [20] **Lu, Y.†**, Seiyon Lee, B., and **Doss-Gollin, J.**: “Bayesian Spatiotemporal Nonstationary Model Quantifies Robust Increases in Daily Extreme Rainfall across the Western Gulf Coast”. *Environmental Research: Climate*. DOI: [10.1088/2752-5295/adf56e](https://doi.org/10.1088/2752-5295/adf56e) 2025
- [19] O'Donnell, M., Murphy, K., **Doss-Gollin, J.**, Dee, S., and Munoz, S. : “Evaluation of Hydroclimatic Biases in the Community Earth System Model (CESM1) within the Mississippi River Basin”. *Hydrology and Earth System Sciences*. DOI: [10.5194/hess-29-4637-2025](https://doi.org/10.5194/hess-29-4637-2025) 2025
- [18] Pollack, A., **Doss-Gollin, J.**, Srikrishnan, V., and Keller, K. : “UNSAFE: An UNcertain Structure and Fragility Ensemble Framework for Property-Level Flood Risk Estimation”. *Journal of Open Source Software*. DOI: [10.21105/joss.07527](https://doi.org/10.21105/joss.07527) 2025
- [17] Kazadi, A., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : “FloodGNN-GRU: A Spatio-Temporal Graph Neural Network for Flood Prediction”. *Environmental Data Science*. DOI: [10.1017/eds.2024.19](https://doi.org/10.1017/eds.2024.19) 2024
- [16] Murphy, K., Dee, S., **Doss-Gollin, J.**, Dunne, K. B. J., O'Donnell, M., and Muñoz, S. : “Competing Influences of Land Use and Greenhouse Gas Emissions on Mississippi River Basin Hydroclimate Simulated over the Last Millennium”. *Paleoceanography and Paleoclimatology*. DOI: [10.1029/2024PA004902](https://doi.org/10.1029/2024PA004902) 2024
- [15] Singh, D., Bekris, Y. S., Rogers, C. D. W., **Doss-Gollin, J.**, Coffel, E. D., and Kalashnikov, D. A. : “Enhanced Solar and Wind Potential during Widespread Temperature Extremes across the U.S. Interconnected Energy Grids”. *Environmental Research Letters*. DOI: [10.1088/1748-9326/ad2e72](https://doi.org/10.1088/1748-9326/ad2e72) 2024

- [14] Amonkar, Y., **Doss-Gollin, J.**, Farnham, D. J., Modi, V., and Lall, U. : “Differential Effects of Climate Change on Average and Peak Demand for Heating and Cooling across the Contiguous USA”. *Communications Earth & Environment*. DOI: [10.1038/s43247-023-01048-1](https://doi.org/10.1038/s43247-023-01048-1) 2023
- [13] Amonkar, Y., **Doss-Gollin, J.**, and Lall, U. : “Compound Climate Risk: Diagnosing Clustered Regional Flooding at Inter-Annual and Longer Time Scales”. *Hydrology*. DOI: [10.3390/hydrology10030067](https://doi.org/10.3390/hydrology10030067) 2023
- [12] **Doss-Gollin, J.**, Amonkar, Y., **Schmeltzer, K.**<sup>†</sup>, and Cohan, D. : “Improving the Representation of Climate Risks in Long-Term Electricity Systems Planning: A Critical Review”. *Current Sustainable/Renewable Energy Reports*. DOI: [10.1007/s40518-023-00224-3](https://doi.org/10.1007/s40518-023-00224-3) 2023
- [11] **Doss-Gollin, J.** and Keller, K. : “A Subjective Bayesian Framework for Synthesizing Deep Uncertainties in Climate Risk Management”. *Earth’s Future*. DOI: [10.1029/2022EF003044](https://doi.org/10.1029/2022EF003044) 2023
- [10] Garcia, M., Juan, A., **Doss-Gollin, J.**, and Bedient, P. : “Leveraging Mesh Modularization to Lower the Computational Cost of Localized Updates to Regional 2D Hydrodynamic Model Outputs”. *Engineering Applications of Computational Fluid Mechanics*. DOI: [10.1080/19942060.2023.2225584](https://doi.org/10.1080/19942060.2023.2225584) 2023
- [9] Wutich, A., Thomson, P., Jepson, W., Stoler, J., Cooperman, A. D., **Doss-Gollin, J.**, Jantrania, A., Mayer, A., Nelson-Núñez, J., Walker, W. S., and Westerhoff, P. : “MAD Water: Integrating Modular, Adaptive, and Decentralized Approaches for Water Security in the Climate Change Era”. *WIREs Water*. DOI: [10.1002/wat2.1680](https://doi.org/10.1002/wat2.1680) 2023
- [8] Zhou, X., Duenas-Osorio, L., **Doss-Gollin, J.**, Liu, L., Stadler, L., and Li, Q. : “Mesoscale Modeling of Distributed Water Systems Enables Policy Search”. *Water Resources Research*. DOI: [10.1029/2022WR033758](https://doi.org/10.1029/2022WR033758) 2023
- [7] **Doss-Gollin, J.**, Farnham, D. J., Lall, U., and Modi, V. : “How Unprecedented Was the February 2021 Texas Cold Snap?”. *Environmental Research Letters*. DOI: [10.1088/1748-9326/ac0278](https://doi.org/10.1088/1748-9326/ac0278) 2021
- [6] **Doss-Gollin, J.**, Farnham, D. J., Ho, M., and Lall, U. : “Adaptation over Fatalism: Leveraging High-Impact Climate Disasters to Boost Societal Resilience”. *Journal of Water Resources Planning and Management*. DOI: [10.1061/\(asce\)wr.1943-5452.0001190](https://doi.org/10.1061/(asce)wr.1943-5452.0001190) 2020
- [5] **Doss-Gollin, J.**, Farnham, D. J., Steinschneider, S., and Lall, U. : “Robust Adaptation to Multiscale Climate Variability”. *Earth’s Future*. DOI: [10.1029/2019ef001154](https://doi.org/10.1029/2019ef001154) 2019
- [4] Rözer, V., Kreibich, H., Schröter, K., Müller, M., Sairam, N., **Doss-Gollin, J.**, Lall, U., and Merz, B. : “Probabilistic Models Significantly Reduce Uncertainty in Hurricane Harvey Pluvial Flood Loss Estimates”. *Earth’s Future*. DOI: [10.1029/2018ef001074](https://doi.org/10.1029/2018ef001074) 2019
- [3] **Doss-Gollin, J.**, Muñoz, Á. G., Mason, S. J., and Pastén, M. : “Heavy Rainfall in Paraguay during the 2015/16 Austral Summer: Causes and Subseasonal-to-Seasonal Predictive Skill”. *Journal of Climate*. DOI: [10.1175/JCLI-D-17-0805.1](https://doi.org/10.1175/JCLI-D-17-0805.1) 2018
- [2] Farnham, D. J., **Doss-Gollin, J.**, and Lall, U. : “Regional Extreme Precipitation Events: Robust Inference from Credibly Simulated GCM Variables”. *Water Resources Research*. DOI: [10.1002/2017WR021318](https://doi.org/10.1002/2017WR021318) 2018
- [1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : “Analytic Modeling of Rainwater Harvesting in the Brazilian Semiarid Northeast”. *Journal of the American Water Resources Association*. DOI: [10.1111/1752-1688.12376](https://doi.org/10.1111/1752-1688.12376) 2015

#### PEER-REVIEWED CONFERENCE PAPERS

- [4] Kazadi, A., **Doss-Gollin, J.**, and Silva, A. : “Pluvial Flood Emulation with Hydraulics-Informed Message Passing”. *Forty-First International Conference on Machine Learning*. URL: <https://openreview.net/forum?id=kIHIA6LrOB> 2024
- [3] Kazadi, A. N., **Doss-Gollin, J.**, Sebastian, A., and Silva, A. : “Flood Prediction with Graph Neural Networks”. *Climate Change AI*. URL: <https://www.climatechange.ai/papers/neurips2022/75> 2022
- [2] Araújo Júnior, L. M., de Souza Filho, F. d. A., da Silva Silveira, C., Aragão Dias, T., and **Doss-Gollin, J.** : “Análise Dos Eventos de Seca No Nordeste Setentrional Brasileiro Com Base No Índice de Precipitação Normalizada”. *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: [10.13140/rg.2.1.4610.7685](https://doi.org/10.13140/rg.2.1.4610.7685) 2014

- [1] **Doss-Gollin, J.**, de Souza Filho, F. d. A., and da Silva, F. O. E. : “Considerações Sobre a Sustentabilidade Hídrica de Cisternas Para Captação de Chuva No Semiárido Brasileiro”. *XII Simpósio de Recursos Hídricos Do Nordeste*. DOI: [10.13140/rg.2.1.4086.4807](https://doi.org/10.13140/rg.2.1.4086.4807) 2014

## SPONSORED RESEARCH

*Amounts reflect Rice portion for collaborative grants and subawards; total amount for direct awards.*

|                                                                                                                                                                                                                                                                                                                                                    |           |
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| PI, National Science Foundation – Infrastructure Systems and People. CAREER: <i>Drain or Retain? Strategic Assessment of Integrated Flood Infrastructure Portfolios with Physics-Informed ML</i> . No. 2543303. (\$599,581)                                                                                                                        | 2026–2031 |
| Co-PI, National Science Foundation – Confronting Hazards, Impacts and Risks for a Resilient Planet (CHIRRP). RAISE: <i>Flood Resilience in Rural Texas Communities</i> PI: Avantika Gori, with Andrew Juan and Keri Stephens. No. 2440167. (\$999,986)                                                                                             | 2025–2028 |
| PI, Consortium for Enhancing Resilience and Catastrophe Modeling (CERCAT). <i>A Nonstationary Joint Probability Method for Tropical Cyclone Hazard Assessment</i> with Avantika Gori. No. P12. (\$75,000)                                                                                                                                          | 2025–2026 |
| PI, Ken Kennedy Institute at Rice University. <i>Advancing AI for Climate Risk and Urban Resilience</i> with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. (\$80,000)                                                                                                                                      | 2025–2026 |
| Co-PI, NVIDIA. <i>Computing Infrastructure for AI-enhanced Climate Risk and Resilience at Rice</i> PI: Arlei Lopes da Silva, with Avantika Gori. in-kind                                                                                                                                                                                           | 2025–2025 |
| PI, Ken Kennedy Institute at Rice University. <i>Advancing AI for Climate Risk and Urban Resilience</i> with Sylvia Dee, Avantika Gori, Arlei Lopes da Silva, Jamie Padgett, and Noemi Vergopolan. (\$80,000)                                                                                                                                      | 2024–2025 |
| Co-PI, National Science Foundation. IUCRC Planning Grant <i>Rice University: Center for Climate, Equity and Resilience in Catmodeling (CERCat)</i> PI: Jamie Padgett, with Katherine Ensor, Avantika Gori, Leonardo Dueñas Osorio, Brian Davison, David Casagrande, Maryam Rahnemoonfar, Paolo Bocchini, and Yi-Chen Yang. No. 2413382. (\$20,000) | 2024–2025 |
| Co-PI, Rice University Sustainability Institute. <i>Workshop on Nature-Based Solutions for Resilient Coastal Cities</i> PI: Philip Bedient, with Avantika Gori. (\$7,500)                                                                                                                                                                          | 2024–2025 |
| PI, Texas Water Development Board. <i>Developing Future Rainfall Frequency Grids for the State of Texas</i> with John Nielsen-Gammon. (Full grant \$192,828. Subaward \$77,750)                                                                                                                                                                    | 2022–2026 |
| Co-PI, National Science Foundation – Climate and Large-Scale Dynamics. <i>Collaborative Research: Evaluating the Past and Future of Mississippi River Hydroclimatology to Constrain Risk via Integrated Climate Modeling, Observations, and Reconstructions</i> PI: Sylvia Dee, with Samuel Muñoz. No. 2147781. (\$472,024)                        | 2022–2025 |
| PI, National Science Foundation – Strengthening America’s Infrastructure. <i>Collaborative Research: EAGER: Participatory Design for Water Quality Monitoring of Highly Decentralized Water Infrastructure Systems</i> with Alicia Cooperman, Alex Mayer, Shane Walker, and Manuela Muñoz Fuerte. No. 2120829. (\$85,046)                          | 2022–2025 |
| Co-PI, 100,000 Strong in the Americas Innovation Fund. <i>IFCE-Rice-SENAI Program on Artificial Intelligence for Urban Sustainability and Resilience to Natural Disasters in the Americas</i> PI: Arlei Lopes da Silva. (\$50,000)                                                                                                                 | 2022–2023 |
| Co-PI, Energy Foundation. <i>Synthesis of Texas Electricity Research from Rice University</i> PI: Daniel Cohan. (\$24,928)                                                                                                                                                                                                                         | 2022–2023 |
| PI, Rice University – Sustainable Futures Fund. <i>Leveraging Earth System Observations at Multiple Scales to Improve Stormwater Management in Houston</i> with Sylvia Dee. (\$50,000)                                                                                                                                                             | 2022–2023 |

## PRESENTATIONS

### INVITED TALKS

|                                                                                                                                                                                                           |      |
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| “Statistical Modeling for Nonstationary Hydroclimate Hazards: Challenges and Opportunities”. Statistics Colloquium, <b>Rice University</b> , Houston, TX.                                                 | 2025 |
| Panelist. Exploring Water Resilience to Emerging Challenges, <b>Society for Risk Analysis</b> , Remote Presentation.                                                                                      | 2025 |
| “Advancing Urban Flood Risk Management through Physics-informed, Data-Driven Hazard Assessment”. Earth, Marine, and Environmental Science Seminar, <b>University of North Carolina</b> , Chapel Hill, NC. | 2024 |

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| "Quantifying and Characterizing Uncertain Climate Hazards to Enable Adaptive Resilience". Atmospheric Sciences Seminar, <b>Texas A&amp;M University</b> , College Station, TX.                                                             | 2022 |
| "Revisiting Our Design Criteria: What Hazards Should We Design for in a Changing Climate?". Hydrologic Sciences and Water Resources Engineering Seminar, <b>University of Colorado Boulder</b> , Remote Presentation.                      | 2022 |
| "Adapting Engineering Design Criteria to a Changing Climate: Insights from House Elevation". Technical Webinar, <b>ASCE Central New Jersey Branch</b> , Remote Presentation.                                                               | 2022 |
| Panelist. Extreme Weather: How To Report on a World That's Warmer, Colder, Wetter, Drier and Weirder, <b>31st Annual Conference of the Society of Environmental Journalists</b> , Houston, TX.                                             | 2022 |
| "Predictable and Preventable: Lessons in Climate Risk Management from the Texas Coast (and Beyond)". Climate Risk and Financial Institutions Seminar (Prof. Madison Condon), <b>Boston University School of Law</b> , Remote Presentation. | 2022 |
| "Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Black-outs". Outreach Event: Science is for Everyone, <b>American Meteorological Society</b> , Remote Presentation.                                   | 2022 |
| "Extreme Impacts Don't Require Extreme Weather: Lessons from the February 2021 Texas Black-outs". Compound Events Working Group, <b>Risk KAN (Knowledge Action Networks)</b> , Remote Presentation.                                        | 2021 |
| Panelist. Tail events: Prediction, Planning, and Performance, <b>Harvard Electricity Policy Group</b> , Remote Presentation.                                                                                                               | 2021 |
| "Towards Adaptive Resilience: Managing Flood Risks in a Changing World". Technical Webinar, <b>ASCE Central New Jersey Branch</b> , Remote Presentation.                                                                                   | 2021 |
| "Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Center for Climate Risk Management CLIMA Seminar, <b>the Pennsylvania State University</b> , State College, PA.                 | 2020 |
| "Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Department of Civil and Environmental Engineering Seminar, <b>Rice University</b> , Houston, TX.                                | 2020 |
| "Prediction and Implications of Structured Climate Risk for Sequential Adaptation under Deep Uncertainty". Complex Systems Simulation and Optimization Group, <b>National Renewable Energy Laboratory</b> , Golden, CO.                    | 2020 |
| "Drivers of Extreme Rainfall: Atmospheric Circulation Patterns and Regional Intense Rainfall in the Ohio River". European Flood Awareness System Group, <b>European Centre for Medium Range Weather Forecasting</b> , Reading, England.    | 2016 |
| "Understanding the Physical Drivers of Extreme Rainfall for Flood Prediction". Oxford Water Network, <b>Oxford University</b> , Oxford, England.                                                                                           | 2016 |

#### FIRST-AUTHOR CONFERENCE AND WORKSHOP PRESENTATIONS

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| "H24A-07: TxRain: A Bayesian framework for integrating historical observations and model projections to develop nonstationary IDF curves". <b>American Geophysical Union Fall Meeting 2025</b> , New Orleans, LA [Talk].                                                                                              | 2025 |
| "Robust Trends in Extreme Rainfall Probabilities in Texas". <b>2025 Texas Climate Conference</b> , <i>Rice University and Texas A&amp;M University</i> , Houston, TX [Talk].                                                                                                                                          | 2025 |
| "Use-Inspired Tools for Climate Hazard Assessment". <b>Nature-Based Solutions for a Resilient Gulf Coast Workshop</b> , <i>Rice University</i> , Houston, TX [Talk].                                                                                                                                                  | 2025 |
| "Advancing Urban Flood Hazard Characterization through Machine Learning: Challenges and Opportunities". <b>American Geophysical Union Fall Meeting 2024</b> , Washington, DC [Talk].                                                                                                                                  | 2024 |
| "Assessing and Managing Climate Risks to Electricity Systems in an Era of Climate Change and Energy Transition". <b>American Geophysical Union Fall Meeting 2024</b> , Washington, DC [Talk].                                                                                                                         | 2024 |
| "Leveraging Machine Learning to Advance Urban Flood Hazard Assessment: Challenges and Opportunities". <b>Data-driven and physics-based machine learning methods for forecasting and knowledge discovery of surface hydrology</b> , <i>Conference on Computational Methods in Water Resources</i> , Tucson, AZ [Talk]. | 2024 |
| "Incorporating Temperature Projections into Energy Systems Planning". <b>Extreme Heat Workshop</b> , <i>Columbia University</i> , New York, NY [Talk].                                                                                                                                                                | 2024 |



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| "NH14B-07: Linking Robust Trends in Observations and Models to Develop Nonstationary Rainfall Frequency Grids for the State of Texas". <b>American Geophysical Union Fall Meeting 2023</b> , San Francisco, CA [Talk].                                                   | 2023 |
| "A Bayesian Spatial Hierarchical Framework for Process-Informed Nonstationary Analysis of Precipitation Frequencies". <b>13th International Workshop on Statistical Hydrology</b> , <i>International Association of Hydrological Sciences</i> , Boston, MA [Talk].       | 2023 |
| "H35F-07: Near-term Predictability Lowers Long-Term Adaptation Costs". <b>American Geophysical Union Fall Meeting 2022</b> , Chicago, IL [Talk].                                                                                                                         | 2022 |
| "Unprecedented Impacts Don't Require Unprecedented Weather". <b>Post-Harvey Climate &amp; Flood Impacts on the Built Environment</b> , <i>Severe Storm Prediction, Education, &amp; Evacuation from Disasters Center</i> , Houston, TX.                                  | 2022 |
| "H25U-1265: Operationalizing Bayesian Model Checking for Robust Decision Making: Insights from House Elevation". <b>American Geophysical Union Fall Meeting 2021</b> , New Orleans, LA [Poster].                                                                         | 2021 |
| "A14H-03: How Unprecedented Was the February 2021 Texas Cold Snap?". <b>American Geophysical Union Fall Meeting 2021</b> , New Orleans, LA [Talk].                                                                                                                       | 2021 |
| "Valuing Flexibility and Soft Instruments for Sequential Decision Problems". <b>2020 Annual Meeting</b> , <i>Society for Decision Making under Deep Uncertainty</i> , [Remote Presentation].                                                                             | 2020 |
| "H11G-07: Towards Adaptive Resilience: Managing Uncertainties and Exploiting Predictability across Timescales". <b>American Geophysical Union Fall Meeting 2019</b> , San Francisco, CA [Talk].                                                                          | 2019 |
| "Adaptive Resilience through Real Options and Deep Reinforcement Learning". <b>Doctoral Consortium on Computational Sustainability</b> , <i>Carnegie Mellon University</i> , Pittsburgh, PA [Talk].                                                                      | 2019 |
| "Evaluating Staged Investments in Critical Infrastructure for Climate Adaptation". <b>2019 Interdisciplinary Ph.D. Workshop in Sustainable Development</b> , <i>Columbia University</i> , New York, NY [Talk].                                                           | 2019 |
| "H52F-05: Robust Adaptation to Cyclical Climate Risk". <b>American Geophysical Union Fall Meeting 2018</b> , Washington, DC [Talk].                                                                                                                                      | 2018 |
| "Robust Adaptation to Multi-Scale Climate Variability". <b>The Nexus of Climate Data, Insurance, and Adaptive Capacity</b> , Asheville, NC [Poster].                                                                                                                     | 2018 |
| "A31H-0135: Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". <b>American Geophysical Union Fall Meeting 2017</b> , New Orleans, LA [Talk].                                                       | 2017 |
| "H22B-02: Designing and Operating Infrastructure for Nonstationary Flood Risk Management". <b>American Geophysical Union Fall Meeting 2017</b> , New Orleans, LA [Talk].                                                                                                 | 2017 |
| "Extreme Rainfall in Paraguay during the 2015-16 Austral Summer: Causes and Predictive Skill". <b>North East Graduate Student Water Symposium</b> , <i>University of Massachusetts Amherst</i> , Amherst, MA [Talk].                                                     | 2017 |
| "Regional Intense Precipitation: Inferences From GCM Atmospheric Circulation Fields". <b>Modeling Research in the Cloud</b> , <i>National Center for Atmospheric Research</i> , Boulder, CO [Poster].                                                                    | 2017 |
| "Statistical-Dynamical Analysis of Climate Projections for Flood Infrastructure Design". <b>Interdisciplinary Ph.D. Workshop in Sustainable Development 2017</b> , <i>Columbia University</i> , New York, NY [Talk].                                                     | 2017 |
| "Causes and Model Skill of the Persistent Intense Rainfall and Flooding in Paraguay during the Austral Summer 2015-2016". <b>Workshop on Subseasonal to Seasonal Predictability of Extreme Weather and Climate</b> , <i>Columbia University</i> , New York, NY [Poster]. | 2016 |
| <b>ADVISEE CONFERENCE AND WORKSHOP PRESENTATIONS</b>                                                                                                                                                                                                                     |      |
| <b>Yuchen Lu<sup>†</sup></b> : "NH13D-0410: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". <b>American Geophysical Union Fall Meeting 2025</b> , New Orleans, LA [Poster].                     | 2025 |
| <b>Yuchen Lu<sup>†</sup></b> : "TxRAIN-Observational: A Hierarchical Bayesian Spatial Framework to Assess Nonstationary Rainfall Intensity, Frequency, and Duration in Texas". <b>American Geophysical Union Fall Meeting 2024</b> , Washington, DC [Talk].              | 2024 |
| <b>Yuchen Lu<sup>†</sup></b> : "Nonstationary Extreme Precipitation Probabilities in Texas". <b>Fall Meeting</b> , <i>Consortium for Enhancing Resilience and Catastrophe Modeling (CERCat)</i> , Bethlehem, PA [Poster].                                                | 2024 |

|                                                                                                                                                                                                                                                                                                                                                    |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Yuchen Lu<sup>†</sup></b> : “Spatially Varying and Duration Dependent Covariate Model: A Hierarchical Bayesian Framework for Multi-duration Extreme Precipitation Frequency Analysis in Texas”. <b>World Environmental &amp; Water Resources Congress 2024</b> , Environmental & Water Resources Institute (EWRI) - ASCE, Milwaukee, WI [Talk]. | 2024 |
| <b>Yuchen Lu<sup>†</sup></b> : “H21T-1602: Spatially Varying Covariate Model: A Hierarchical Bayesian Framework for Precipitation Frequency Analysis in the Gulf Coast”. <b>American Geophysical Union Fall Meeting 2023</b> , San Francisco, CA [Talk].                                                                                           | 2023 |
| <b>Yuchen Lu<sup>†</sup></b> : “H42E-1333: Nonstationary GEV with Hierarchical Spatial Pooling: A Spatiotemporal Bayesian Framework for Nonstationary Extreme Precipitation Frequency Analysis in the Gulf Coast”. <b>American Geophysical Union Fall Meeting 2022</b> , Chicago, IL [Talk].                                                       | 2022 |

## TEACHING AND ADVISING

### COURSES TAUGHT

|                                                                                                 |             |
|-------------------------------------------------------------------------------------------------|-------------|
| Rice CEVE 421/521: <i>Climate Risk Management</i> .                                             | Spring 2026 |
| Rice CEVE 543: <i>Statistical-Physical Methods for Hydroclimate Extremes and Catastrophes</i> . | Fall 2025   |
| Rice CEVE 101: <i>Fundamentals of Civil and Environmental Engineering</i> .                     | Fall 2024   |
| Rice CEVE 421/521: <i>Climate Risk Management</i> .                                             | Spring 2024 |
| Rice CEVE 543: <i>Data Science Methods for Climate Hazard Assessment</i> .                      | Fall 2023   |
| Rice CEVE 421/521: <i>Climate Risk Management</i> .                                             | Spring 2023 |
| Rice CEVE 543: <i>Environmental Data Science</i> .                                              | Spring 2022 |
| Rice CEVE 101: <i>Fundamentals of Civil and Environmental Engineering</i> .                     | Fall 2021   |

### CURRENT GRADUATE STUDENTS

|                                                                                                                         |       |
|-------------------------------------------------------------------------------------------------------------------------|-------|
| True Furrh: <b>Ph.D.</b> in Civil and Environmental Engineering, Rice University. <b>Committee Member</b> .             | 2021– |
| Dongwook Kim: <b>Ph.D.</b> in Civil and Environmental Engineering, Rice University. <b>Primary Advisor</b> .            | 2024– |
| Yuchen Lu: <b>Ph.D.</b> in Civil and Environmental Engineering, Rice University. <b>Primary Advisor</b> .               | 2021– |
| Kelsey Murphy: <b>Ph.D.</b> in Earth, Environmental, and Planetary Sciences, Rice University. <b>Committee Member</b> . | 2021– |
| Jiayue Yin: <b>Ph.D.</b> in Earth, Environmental, and Planetary Sciences, Rice University. <b>Committee Member</b> .    | 2023– |

### PAST GRADUATE STUDENTS

**Primary Advisor:** Katlyn Schmeltzer, MCEE 2025.

**Committee Member:** Valeriia Sobolevskaia, Ph.D. 2026; Karan Jakhar, Ph.D. 2025; Kyle Ostlind, M.S. 2025; John A. Baer, M.S. 2024; Kendall Capshaw, Ph.D. 2024; Xinyue Luo, Ph.D. 2024; Anibal Tafur Gutierrez, Ph.D. 2024; Matthew Garcia, Ph.D. 2023; Mia Peebles, M.S. 2023; Xiangnan Zhou, Ph.D. 2023; Raychel Bahnick, M.S. 2022; Alyssa Graham, M.S. 2022; Elizabeth Hoffmann, M.S. 2022; Chunshan Liu, Ph.D. 2022; Xiaoyu (Toby) Li, M.S. 2021.

### UNDERGRADUATE RESEARCHERS

*Year indicates graduation year.*

|                                                                                    |      |
|------------------------------------------------------------------------------------|------|
| Zain Rahman: <i>B.S. in Computer Science</i> , Rice University.                    | 2027 |
| Kyle Olcott: <i>B.S. in Civil and Environmental Engineering</i> , Rice University. | 2025 |
| Sophia Prieto: <i>B.S. in Statistics</i> , Rice University.                        | 2023 |
| John Cook: <i>B.S. in Civil and Environmental Engineering</i> , Rice University.   | 2022 |

### TEAMS ADVISED

|                                                                                                                                                                          |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Optimal Policy for Decentralized Wastewater Systems (DWS) while Relaxing Certainty:</b> Rice Computational Mathematics and Operations Research (CMOR) Senior Project. | 2024-2025   |
| <b>Flood Sight   Advancing Real-Time Flood Predictions for Situational Awareness:</b> Rice Data to Knowledge (D2K) Lab.                                                  | Spring 2025 |
| <b>Predicting Streamflow for Hill Country Gauges Using a Transformer-based Model:</b> Rice Data to Knowledge (D2K) Lab.                                                  | Fall 2025   |

## SERVICE ACTIVITIES

### DEPARTMENTAL SERVICE

|                                                             |           |
|-------------------------------------------------------------|-----------|
| <b>Member</b> , Graduate Studies Committee.                 | 2024–     |
| <b>Member</b> , Diversity, Equity, and Inclusion Committee. | 2024      |
| <b>Member</b> , Faculty Search Committee.                   | 2022–2023 |
| <b>Member</b> , Seminar Committee.                          | 2022–2023 |

### UNIVERSITY SERVICE

|                                                                                                       |           |
|-------------------------------------------------------------------------------------------------------|-----------|
| <b>Member</b> , Research Council. Ken Kennedy Institute.                                              | 2024–     |
| <b>Faculty Associate</b> . Duncan College.                                                            | 2023–     |
| <b>External Search Committee Member</b> . Department of Earth, Environmental, and Planetary Sciences. | 2022–2023 |

### PROFESSIONAL SERVICE

|                                                                                                                                                                                    |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Research Council Member</b> . Ken Kennedy Institute.                                                                                                                            | 2024– |
| <b>Committee Member</b> , Water and Society Newsletter Committee. American Geophysical Union.                                                                                      | 2024– |
| <b>Committee Member</b> , Environmental and Water Resources Systems (EWRS) Committee. American Society of Civil Engineers (ASCE) Environmental & Water Resources Institute (EWRI). | 2021– |

### PEER REVIEW

**Journals:** AGU Advances; Climate Risk Management; Climatic Change; Communications Earth and Environment; Earth's Future; Energy Technology; Engineering Applications of Computational Fluid Mechanics; Environmental Data Science; Environmental Research Letters; Environmental Research: Climate; Geophysical Research Letters; Hydrology and Earth System Sciences; IEEE Transactions on Geoscience and Remote Sensing; Joule; Journal of Applied Meteorology and Climatology; Journal of Hydrology; Journal of Water Resources Planning and Management; Machine Learning: Earth; Natural Hazards and Earth System Sciences; npj Natural Hazards; Oxford Development Studies; Water Resources Research; Water Security; Weather, Climate, and Society.

**Funding Agencies:** Department of Energy (BER); Dutch Research Council (NWO); National Science Foundation.

**Other:** Electric Power Research Institute (EPRI); Texas Water Development Board (TWDB).

### SESSIONS CONVENED

|                                                                                                                                                                                                                                                            |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Co-Organizer</b> . <i>Nature-Based Solutions for a Resilient Gulf Coast</i> . <b>Rice University</b> , Houston, TX.                                                                                                                                     | 2025 |
| <b>Primary Convener</b> . <i>H31G - Integrating Social, Scientific, and Engineering Approaches to Identify and Address Gaps in Water Infrastructure and Household Water Security</i> . <b>American Geophysical Union Fall Meeting</b> , San Francisco, CA. | 2023 |
| <b>Convener</b> . <i>NH41C - Hybrid Modeling and Digital Twin Systems for Flood Hazard Prediction and Risk Assessment at Different Spatial Scales</i> . <b>American Geophysical Union Fall Meeting</b> , Washington, DC.                                   | 2023 |
| <b>Chair</b> . <i>H44G - Water and Society: Interdisciplinary Perspectives on Hydroclimatic Forecasting for Water Resources Decision Making</i> . <b>American Geophysical Union Fall Meeting</b> , New Orleans, LA.                                        | 2021 |
| <b>Primary Convener</b> . <i>NH53 - Emerging Needs and Approaches for Climate Services: Understanding and Developing Innovative Approaches to User-Oriented Climate Services</i> . <b>American Geophysical Union Fall Meeting</b> , San Francisco, CA.     | 2019 |
| <b>Student Organizer</b> . <i>Earth and Environmental Engineering Student Research Symposium</i> . <b>Columbia University</b> , New York, NY.                                                                                                              | 2018 |
| <b>Student Organizer</b> . <i>Earth and Environmental Engineering Student Research Symposium</i> . <b>Columbia University</b> , New York, NY.                                                                                                              | 2017 |

### MEDIA HIGHLIGHTS

|                                                                                                                                                     |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------|
| "FEMA elabora nuevos mapas de riesgo de inundación ¿Afectará esto tu propiedad?". <b>Telemundo Houston</b> [recorded video interview].              | 2026 |
| "The Climate-Disaster Scores That Could Make or Break Your Home Sale". <b>The Wall Street Journal</b> – Jean Eaglesham and Nicole Friedman [print]. | 2026 |
| "Historic Texas Flooding". <b>KEYE-AUS (CBS)</b> [recorded video interview].                                                                        | 2025 |



|                                                                                                                                                                                                                                                                    |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| "Houston's Morning Show". <b>Fox26 Houston</b> [live video interview].                                                                                                                                                                                             | 2025 |
| "Breaking down the force of water in the Texas floods". <b>AP News</b> — Michael Phillis [print].                                                                                                                                                                  | 2025 |
| "Here are some things you can do to be better prepared for major flooding". <b>Associated Press</b> — Caleigh Wells [print].                                                                                                                                       | 2025 |
| "ABC13-Luke Jones speaks with James Doss-Gollin, an assistant professor of Civil and Environmental Engineering at Rice University, about the key differences between flooding in Houston and central Texas." <b>ABC13</b> — Luke Jones [recorded video interview]. | 2025 |
| "After deadly flooding in Central Texas, state lawmakers look to prevent similar tragedies". <b>KVUE</b> — Daniel Perreault [recorded video interview].                                                                                                            | 2025 |
| "New Flood Warning System Greenlit Shortly Before Deadly Texas Disaster". <b>The Wall Street Journal</b> — Joseph De Avila [print].                                                                                                                                | 2025 |
| "Questions arise on how emergency warning systems work after Central Texas flood". <b>ABC13</b> — Tom Abrahams [print].                                                                                                                                            | 2025 |
| "Bajo la Amenaza del Golfo (Under the Threat of the Gulf)". <b>Telemundo Houston</b> [recorded video interview].                                                                                                                                                   | 2025 |
| "Will Texas become too hot for humans?". <b>BBC Future</b> — Sarah Griffiths [print].                                                                                                                                                                              | 2023 |
| "Climate change has sent temperatures soaring in Texas". <b>The Texas Tribune</b> — Erin Douglas, Yuriko Schumacher and Alex Ford [print].                                                                                                                         | 2023 |
| "Texas could have foreseen 2021 cold-wave disaster, new study concludes". <b>Texas Climate News</b> — Bob Henson [print].                                                                                                                                          | 2022 |
| "Opinion: The risks of climate change are great - so are the rewards of solving it". <b>Houston Chronicle</b> — Andrew Dessler, James Doss-Gollin, and Katherine Hayhoe [print].                                                                                   | 2021 |
| "The False Comfort of Higher Seawalls". <b>The New Republic</b> — Paola Rosa-Aquino [print].                                                                                                                                                                       | 2019 |
| "New Study Shows Promise for Long-Term Weather Forecasts in South America". <b>State of the Planet</b> — Elisabeth Gawthrop [print].                                                                                                                               | 2018 |

## LANGUAGE SKILLS

English: Native  
 Spanish: Fluent  
 Portuguese: Proficient

## TRAINING AND EXPERIENCE

|                                                                                                                                                                                        |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Strategic Program to Accelerate Researchers in Computing (SPARC) Participant</b> — <i>Natural Hazards Engineering Research Infrastructure (NHERI) DesignSafe-Cl.</i>                | 2025      |
| <b>Panel Fellow</b> — <i>NSF CMMI's Game Changer Academies for Advancing Research Innovation.</i>                                                                                      | 2021      |
| <b>Visiting Graduate Researcher</b> — <i>Lamontagne Research Group, Department of Civil and Environmental Engineering, Tufts University, Medford, MA.</i>                              | 2019–2020 |
| <b>Summer School Participant</b> — <i>Fluid Dynamics of Sustainability and the Environment, Cambridge University, Cambridge, England.</i>                                              | 2016      |
| <b>Education Policy Intern</b> — <i>Elm City Communities / New Haven Housing Authority, New Haven, CT.</i>                                                                             | 2015      |
| <b>President (2014), Design Lead (2013), Member (2012, 2015)</b> — <i>Engineers Without Borders, Yale Student Chapter, New Haven, CT.</i>                                              | 2012–2015 |
| <b>Undergraduate Research Assistant</b> — <i>Lab of Jaehong Kim, Department of Chemical and Environmental Engineering, Yale University, New Haven, CT.</i>                             | 2014–2015 |
| <b>Visiting Undergraduate Researcher</b> — <i>Water and Climate Risk Lab, Department of Hydraulic and Environmental Engineering, Universidade Federal do Ceará, Fortaleza, Brazil.</i> | 2014      |
| <b>Mechanical Design Intern</b> — <i>Slingshot Team, DEKA Research &amp; Development, Manchester, NH.</i>                                                                              | 2013      |
| <b>Undergraduate Research Assistant</b> — <i>Lab of Jan Schroers, Department of Mechanical Engineering and Materials Science, Yale University, New Haven, CT.</i>                      | 2012      |
| <b>Ikatú Agua Intern</b> — <i>Fundación Paraguaya, Asunción, Paraguay.</i>                                                                                                             | 2012      |